Derek Yao

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Education

University of Washington, Paul G. Allen School of Computer Science — Seattle, WA

Bachelor of Science in Computer Science | Dean's List | GPA: 3.89 — *Expected Jun 2028*

Coursework: Data Structures and Parallelism, System/Software Design, Advanced Linear Algebra, Multivariable Calculus, Hardware/Software Interface, Discrete Math, Differential Equations, Natural Language Processing, Artificial Intelligence

Experience

Contour (Backed by Lightspeed) — Jun 2026 – Present

Software Engineering Intern, Employee #5

- Shipped real-time bidirectional Linear sync processing **20K+** task statuses across **12+** live customer deployments
- Scaled meeting bot infra to **120K webhooks/hr** with concurrency, turn-level streaming, and calendar syncs
- Improved Memgraph retrieval latency **7.8x** while preserving **94%** recall through query and hydration optimization
- Automated FDE template creation by building an agent-native spreadsheet editor, XML parser, and tool interface

University of Washington Denolle Lab — Dec 2025 – Present

Undergraduate Researcher

- Built **distributed** CNN inference system for seismic event detection across sparse Alaskan edge sensor network
- Shrank memory footprint **92%** through distillation and **INT8** quantization, enabling inference on edge hardware
- Reduced inference latency **4.4x** (420 to 65ms) with pruning and operator fusion while preserving **0.96 F1**

Oracle — Dec 2025 – Apr 2026

Product Manager – Contract

- Advised Oracle Fusion Cloud SCM leadership on enterprise agent adoption across logistics/transportation workflows
- Designed AI implementation strategy by identifying workflow traceability, data-lineage, and integration gaps

University of Washington GAIA-HazLab — Oct 2025 – Present

Undergraduate Researcher

- Unified **500+** fragmented papers into citation-grounded RAG knowledge base for **22** earth science researchers
- Achieved **92% Recall@K** by fusing semantic search with OCR metadata extraction and indexing of scientific figures
- Built research agent that surfaces papers for cross-domain queries, reducing literature search from **mins to seconds**

Projects

SpeakEasy | *FastAPI, React, Fish Audio, OpenFace, RetinaFace, Claude, SSE* — Dec 2025

- Won **Honorable Mention @CalHacks** by building live speaker coaching tool fusing gaze, facial and audio signals
- Cut feedback latency to **<300ms** by decoupling heavier multimodal inference with asynchronous SSE pipelines
- Improved real-time attention classification accuracy to **91%** using geometric gaze estimation and face tracking

Forklift | *Python, FastAPI, GitHub API, OpenAI API* — June 2025

- Won **Best Hack + $1,200 @Roo Code Hackathon** by building better issue triage and onboarding for open-source
- Utilized Github API and hybrid RRF of **BM25** + vector search to surface context-aware potential contributions
- Helped onboard **200+ developers** onto various open-source projects through context-aware issue triage

Persona-GPT | *Python, PyTorch, CUDA, tiktoken* — Aug 2026

- Built a **6-layer GPT** from scratch with multi-head causal attention, residual blocks, and autoregressive decoding
- Experimented with **persona drift** in conversation using proxies for sentiment, verbosity, and lexical diversity
- Probed **reward hacking** by post-training against conversational proxies and measuring unintended behavioral shifts

Technical Skills

Programming Languages: Python, Golang, Java, JavaScript, TypeScript, SQL, C, x86 Assembly
Frameworks: Django, FastAPI, React, Node.js, TensorFlow, OpenCV, PyTorch, NumPy, MCP, RAG
Developer Tools: AWS, Postgres, Git, Supabase, SQLite, Kubernetes, Linux, Docker, Oban, Redis, Memgraph, Temporal
Concepts: DSA, OOP, NLP, System Design, Distributed Systems, RESTful APIs, CI/CD, Database Systems, Vector Search, Embeddings, Data Modeling, Event-Driven Architecture, Async Processing, Machine Learning, Agent Orchestration